

SEERAJA AJAY KONNUR

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SUMMARY

AI Scientist building intelligent systems at the intersection of machine learning and computational biology to accelerate scientific discovery and advance data-driven healthcare.

Research Interests: AI for Science • Foundation Models • Biological Sequence Modelling • Computational Biology • Evolutionary Modelling

EDUCATION

King's College London

London, United Kingdom

MSc Applied Statistical Modelling and Health Informatics

September 2021 – September 2022

Relevant Coursework: Machine learning, Artificial Intelligence (AI), Natural language processing, Health Informatics, Statistical Modelling, Prediction Modelling, Computational Neuroscience

Institute of Bioinformatics and Applied Biotechnology

Bangalore, India

MSc Bioinformatics and Applied Biotechnology

July 2017 - May 2019

Relevant Coursework: Bioinformatics, Molecular and Cellular Biology, Computer Science, Mathematics and Statistics

Ramnarain Ruia Autonomous College

Mumbai, India

BSc Microbiology

June 2014 - April 2017

Subjects: Microbiology, Chemistry, Life Sciences

SKILLS

Programming: Python, R, C++, Bash scripting (Linux OS), MATLAB

AI & Machine Learning: Foundation Models, Transformers, Graph Neural Networks, Statistical & Predictive Modelling

Libraries & Frameworks: PyTorch, TensorFlow, Scikit-learn, Pandas, NumPy, PyTorch Geometric, Matplotlib

Tools & Technologies: Git, Docker, AWS (S3, EC2), MySQL, Jira

WORK EXPERIENCE

ARTPARK @ IISc

Bangalore, India

Associate Machine Learning Scientist - Computational Biology

August 2024 – Present

- Spearheading AI and computational biology research across genomics, disease surveillance and outbreak intelligence.
- Developed a novel AI-driven framework for influenza reassortment prediction by combining foundation models with biologically informed genetic algorithms; accepted at **NeurIPS 2025** FM4LS Workshop. [[Paper](#)] [[Github](#)]
- Building a sequence-based vaccine matching framework using epitope analysis and protein language models.
- Driving interdisciplinary collaborations across AI, biology, and public health.
- Mentoring junior researchers through the end-to-end project lifecycle, from research design to shipping production-ready computational tools.

AarogyaAI Innovations Pvt. Ltd.

Remote

Bioinformatics Engineer

June 2023 – December 2023

- Developed machine learning models for antimicrobial drug resistance prediction across 4 major pathogens.
- Designed novel ML architectures by integrating bioinformatics pipelines with cutting edge machine learning models resulting in 10% improvement in prediction accuracy.
- Collaborated with cross-functional teams and led my team of 5 members as a SCRUM Master to successfully release the product.

- Utilized machine learning on multimodal patient data to classify oral cancer tissue samples.
- Achieved vertical integration by combining 3 types of omics data including transcriptomics and epigenomics.
- Reduced high-dimensional data (over 50,000 features) to 500 using statistical data reduction and deep learning (stacked autoencoders) techniques. [[Github](#)]

PUBLICATIONS / CONFERENCES

Paper 'Predicting Influenza A Reassortment Potential Using Foundation Models and Genetic Algorithms for Pandemic Preparedness' accepted at **NeurIPS 2025** Workshop on Multi-modal Foundation Models and Large Language Models for Life Sciences (FM4LS). [[Paper](#)] [[Github](#)]

INVITED TALKS

Invited Speaker, ICAR – Indian Veterinary Research Institute (IVRI), Bangalore

Machine Learning Approaches for Next-Generation Vaccine Development

ACADEMIC SERVICE

Reviewer, Multi-modal Foundation Models and Large Language Models for Life Sciences Workshop ICML 2026

Reviewer, Multi-modal Foundation Models and Large Language Models for Life Sciences Workshop NeurIPS 2025

Reviewer, Multi-modal Foundation Models and Large Language Models for Life Sciences Workshop ICML 2025

PROJECTS

Cardiovascular Disease Risk Prediction Model

- Built a predictive model using multimodal data for early detection and prevention of cardiovascular diseases.
- Performed predictive analytics on key clinical factors and pulse waveform data of over 1,27,000 individuals from the UK Biobank.
- Optimized and evaluated the model, achieving an AUC of 0.77 [[Github](#)].

Host Pathogen Protein – Protein Interaction Network

- Enhanced a Host–Pathogen Protein–Protein Interaction Network (PPIN) using recommender systems (probabilistic spreading algorithm) and supervised machine learning algorithms.
- Identified 35 new interactions providing valuable insights into potential drug targets. [[Github](#)]

Deep Learning for Speech Samples Classification

- Investigated speech samples using convolutional neural networks to detect the presence of cold and flu. [[Github](#)]

Computational Neuroscience

- Analyzed 3D brain imaging data using unsupervised machine learning algorithms to differentiate images into distinct clusters.
- Delivered a cutting-edge brain age prediction model by harnessing functional connectivity data from resting-state functional MRI through supervised machine learning methods.

Systems Biology Computational Analysis of Pancreatic Beta Cells

- Implemented the published model of Pancreatic Beta cell on MATLAB and contributed to open source on ModelDB.
- Modelled various molecules and pathways yielding new insights into molecular and cellular biology. [[ModelDB](#)]